

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 5_COD_Question 2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Mike is learning about Binary Search Trees (BSTs) and wants to implement various operations on them. He wants to write a basic program for creating a BST, inserting nodes, and printing the tree in the pre-order traversal.

Write a program to help him solve this program.

Input Format

The first line of input consists of an integer N, representing the number of values to insert into the BST.

The second line consists of N space-separated integers, representing the values to insert into the BST.

Output Format

The output prints the space-separated values of the BST in the pre-order traversal.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

3 1 5 2 4

Output: 3 1 2 5 4

Answer

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
};
```

```
struct Node* createNode(int value) {  
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));  
    newNode->data = value;  
    newNode->left = newNode->right = NULL;  
    return newNode;  
}
```

```
// Function to insert a node into the BST  
struct Node* insert(struct Node* root, int value) {  
    if (root == NULL) {  
        return createNode(value);  
    }  
  
    if (value < root->data) {  
        root->left = insert(root->left, value);  
    } else {  
        root->right = insert(root->right, value);  
    }  
}
```

```
        return root;
    }

    // Function to perform pre-order traversal
    void printPreorder(struct Node* root) {
        if (root != NULL) {
            printf("%d ", root->data);
            printPreorder(root->left);
            printPreorder(root->right);
        }
    }

    int main() {
        struct Node* root = NULL;

        int n;
        scanf("%d", &n);

        for (int i = 0; i < n; i++) {
            int value;
            scanf("%d", &value);
            root = insert(root, value);
        }

        printPreorder(root);
        return 0;
    }
```

Status : Correct

Marks : 10/10